DOD-P-85321(AS)
16 May 1980
METRIC

MILITARY SPECIFICATION

FOR

PROPYLENE OXIDE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one grade of propylene oxide.

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6830

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 268-74	Sampling and Testing Volatile Solvents for Use in Paint, Varnish, Lacquer, and Related Products, Methods of.
ASTM D 1298-67	Density, Specific Gravity, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method, Test Method for (API STD 2547) (R 1977) (Redesignated ANSI/ASTM D 1298).
ASTM D 1364-64	Water in Volatile Solvents (Fischer Reagent Titration Method) Method of Test for (R 1976).
ASTM D 2108-71	Color of Halogenated Organic Solvents and Their Mixtures (Platinum-Cobalt Scale) Method of Test for (R 1978).
ASTM D 2111-71	Specific Gravity of Halogenated Organic Solvents and Their Mixtures, Standard Methods of Test for (R 1978).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS.

3.1 General material requirements. The propylene oxide shall meet the following product characteristics and quality.

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3.1.1 Chemical properties. The chemical properties of the propylene oxide shall be as specified in Table I.

TABLE I. Chemical properties.

Property	Specification limits	
	Minimum	Maximum
Assay (C_3H_6O)	99.0%	----
Specific gravity 20°C/20°C 25°C/25°C	 0.829 0.825	 0.831 0.827
Water	----	0.05%
Acidity (as acetic acid)	----	0.005%
Chloride, inorganic	----	0.005%
Nonvolatile residue	----	0.010g/100ml
Color (APHA)	----	15 1/
Suspended matter	Substantially free	
Organic impurities:		
Vinyl chloride	----	Trace
Vinylidene chloride	----	Trace
Acetaldehyde	----	0.005%
Ethylene oxide	----	Trace
Propionaldehyde	----	0.005%
1,2-Dichloropropane	----	Trace
Propylene glycol	----	Trace

1/ APHA Pt-Co Scale

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3.1.2 Workmanship. Propylene oxide shall be free from foreign materials, rust, and contamination and shall be uniform in quality.

3.2 First article (preproduction) sample. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor shall furnish a sample of propylene oxide for first article inspection and approval (see 4.2.1). The preproduction sample shall be manufactured using the same methods, materials, processes, and procedures proposed for production. Any production prior to acceptance of the preproduction sample is at the risk of the contractor.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Special tests and examinations.

4.2.1 First article (preproduction) inspection. The preproduction sample consisting of not less than 1 kilogram shall be subjected to all the tests specified herein in any sequence desired. Failure of the preproduction sample to meet any of the requirements specified herein shall reject the preproduction sample and the batch from which the sample was taken.

4.3 Quality conformance inspections. The test conditions of 4.4 shall be used to determine that the requirements of this specification have been met. Unless otherwise specified herein, all tests shall be performed in duplicate. The average of the two results shall be taken as the test result.

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4.3.1 Inspection lot. A lot shall consist of material produced at one plant during continuous production with no change in formulation or process and offered for acceptance at one time. If manufacture is by batch (see 6.3.1) process, then a lot shall consist of not more than one batch.

4.3.2 Acceptance sampling. A sample of not less than 1 kilogram, representative of the lot, shall be selected from each lot. Unless otherwise specified, the sample shall be composed of equal amounts from each container which constitutes a lot of material. In those cases where a lot shipment will consist of more than 10 containers, not less than 10 containers shall be selected at random from which to obtain the sample. The sample shall be submitted for testing as specifed in the contract. Failure of the sample to meet any of the requirements specified herein shall reject the lot represented.

4.4 Test conditions. Unless otherwise specified herein, the following test conditions shall apply:

(a) Temperature: Between 15 and 35 degrees Celsius (°C).

(b) Altitude: Normal ground elevation of test facility.

(c) Humidity: Not to exceed 90 percent relative.

4.4.1 Test methods.

4.4.1.1 Assay (C_3H_6O). An assay of the propylene oxide shall be made as follows:

4.4.1.1.1 Reagents. Reagents used shall be as follows:

(a) Magnesium chloride ($Mg\ Cl_2 \cdot 6H_2O$)

(b) Saturated magnesium chloride in water (approximately 65 percent)

(c) Sodium hydroxide, 1 Normal (N) (standardized)

(d) Sulfuric acid, 1N (standardized)

(e) Methyl red, 0.1 percent, in a 1:1 mixture of methanol and water.

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4.4.1.1.2 Apparatus. Apparatus used shall be as follows:

(a) Glass ampoules, 2 milliliter (ml) volume

(b) Pressure bottles, (350 ml approximately), narrow mouth

(c) Magnetic stirrer and a polytetrafluoroethylene (PTFE) coated stirring bar

(d) Burette, 10 ml volume.

4.4.1.1.3 Procedure. Tests shall be performed as follows:

(a) Weigh a dry, clean ampoule to the nearest 0.1 milligram (mg). Add 1.00 ±0.05 grams (gm) of sample using a syringe or capillary pipette. Draw out the tip of the ampoule in a flame to seal. Reweigh when cooled, including the drawn-off tip of the ampoule.

(b) Place 100 gm of magnesium chloride, 85 ml of saturated magnesium chloride solution, and 20 ml of 1N H_2SO_4, into a pressure bottle. (Keep ratio of 20 milliequivalents of acid to 1 gm of sample. More magnesium chloride is needed with an increased amount of acid.) Introduce the sealed ampoule into the bottle and stopper tightly, taking care not to break the ampoule. Cool in an ice bath for 15 minutes.

(c) Remove the bottle, wrap with a towel and shake to break the ampoule. Allow the contents to react, shaking intermittently for 15 minutes. Then shatter the ampoule completely with a stout glass rod.

(d) Add 5 drops of methyl red indicator solution and titrate the excess acid with 1N sodium hydroxide. Run a blank titration on the reagents, as in (b), using exactly the same amount of sulfuric acid. Add sodium hydroxide very slowly with very rapid stirring, especially at the approach of the end point (pink fading to light yellow). Check end point by adding 1 drop of 1N sulfuric acid. If the solution does not change back to pink, the end point has been exceeded. Titration will be about 1 ml.

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4.4.1.1.4 Calculation. The following calculation shall be made:

$$\text{Percent propylene oxide} = \frac{(B - A)\ N \times 5.808}{\text{gm of sample}}$$

where:

A = volume (milliliters) of sodium hydroxide solution required for the sample

B = volume (milliliters) of sodium hydroxide required for the blank

N = normality of sodium hydroxide solution

To be acceptable, the assay shall be no less than the value specified in Table I.

4.4.1.2 Specific gravity. The specific gravity of propylene oxide at either 20°C/20°C or 25°C/25°C to ±0.001 unit (see Table I) shall be determined by one of the following methods:

(a) Pycnometer method, as specified in ASTM D 2111-71, Method C.

(b) Hydrometer method, as specified in ASTM D 1298-67 or ASTM D 2111-71, Method B.

To be acceptable, the specific gravity shall meet the requirements of Table I.

4.4.1.3 Water. The water content of the propylene oxide shall be determined by the Karl Fischer method, as specified in ASTM D-1364, using methanol as a solvent, and either a visual or electrometric end point. To be acceptable, the water content shall be no greater than the value specified in Table I.

4.4.1.4 Acidity (as acetic acid). The acetic acid content of the propylene oxide shall be determined as follows:

4.4.1.4.1 Reagents. Reagents used shall be as follows:

(a) Aqueous sodium hydroxide 0.01N (standardized)

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(b) Bromothymol blue indicator (pH range 6.2-7.6) 0.1 percent in water.

4.4.1.4.2 Apparatus. Apparatus used shall be as follows:

(a) Burette, 10 ml

(b) Titration vessel (250 ml Erlenmeyer is suitable)

(c) Magnetic stirrer and PTFE-coated stirring bar.

4.4.1.4.3 Procedure. Tests shall be performed as follows:

(a) Add a known mass or volume of approximately 50 gm or 60 ml of propylene oxide into a 250 ml Erlenmeyer flask. Add four drops of indicator and titrate to a blue end point, stirring rapidly and adding the sodium hydroxide slowly as it goes into the greenish-blue transition state.

4.4.1.4.4 Calculation. The following calculation shall be made:

$$\text{Percent acidity as acetic acid} = \frac{V \times N \times 6.0}{B \times G} \text{ or } = \frac{V \times N \times 6.0}{M}$$

where:

V = volume (milliliters) of sodium hydroxide titration

N = normality gram equivalent mass per liter (eq/l) of sodium hydroxide

M = mass (grams) propylene oxide

B = volume (milliliters) of propylene oxide

G = specific gravity of sample.

To be acceptable, the acidity shall be no greater than the value specified in Table I.

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4.4.1.5 Chlorides, inorganic.

4.4.1.5.1 Reagents. Reagents used shall be as follows:

(a) Nitric acid concentrated

(b) Silver nitrate approximately 0.1N in water (standardization not required)

(c) Chloride standard solution, 100 mg chloride ion per liter.

4.4.1.5.2 Apparatus. Apparatus used shall be as follows:

(a) Graduated cylinders, 100 ml (glass stoppered) or Nessler tubes, 100 ml, or equivalent

(b) Turbidimeter or nephelometer (optional).

4.4.1.5.3 Procedure. The inorganic chlorides shall be determined as follows:

(a) Place 40 ml of distilled water in each of two glass cylinders or tubes and cool in an ice bath. Add 12 ml of propylene oxide to one cylinder and 5 ml of chloride ion standard to the other cylinder and dilute to 50 ml with distilled water. Add 10 drops of concentrated nitric acid and 10 ml of 0.1N silver nitrate solution to each. Mix and compare the turbidity or precipitation formed visually in the tubes or by turbidimeter. If turbidity is equal to or less than that of the chloride ion standard, the sample is acceptable.

NOTE: This is equivalent to not greater than 50 parts per million chloride in the propylene oxide sample. If turbidity is greater than that of the chloride ion standard, proceed to (b) below.

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(b) If turbidity is formed, make up the above solutions again and prepare an additional blank in the same type of container by cooling 50 ml of distilled water in an ice bath and adding 12 ml of propylene oxide. Add 10 drops of nitric acid and compare the turbidity of this blank to that of the solution containing silver nitrate. Inorganic chloride is present only if the solution containing silver nitrate is more turbid than this blank. To be acceptable, the inorganic chloride content shall be no greater than the value specified in Table I.

4.4.1.6 Nonvolatile residue.

WARNING

Propylene oxide boils at 34.3°C and the vapors are extremely flammable. Perform this test under a forced-draft exhaust hood away from flames, sparks, or other sources of ignition.

4.4.1.6.1 Apparatus. Apparatus used shall be as follows:

(a) 100 ml graduated cylinder

(b) Glass or porcelain evaporating dish of about 200 ml capacity

(c) Electric oven (100°C-110°C).

4.4.1.6.2 Procedures. Tests shall be performed as follows:

(a) Clean the evaporating dish, dry in 100°C-110°C oven, cool in desiccator, and weigh to 0.0001 gm.

(b) Add a known volume (approximately 100 ml) or mass (approximately 80 gm) of the sample to the dish.

(c) Place the vessel and sample on a clean, dry surface under the exhaust hood with a good air flow over the sample.

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(d) When the sample has evaporated completely dry, place the vessel in the 100°C-110°C oven for 1 hour. Cool in a desiccator and reweigh.

4.4.1.6.3 Calculation. The following calculation shall be made:

$$\text{Nonvolatile residue, percent by mass} = \frac{(A - B) \times 100}{V \times G} \text{ or } = \frac{(A - B) \times 100)}{M}$$

where:

A = mass (grams) of vessel plus nonvolatile residue

B = mass (grams) of vessel

G = specific gravity of sample

M = mass (grams) of sample

V = volume (milliliters) of propylene oxide sample.

To be acceptable, the nonvolatile residue shall be no greater than the value specified in Table I.

4.4.1.7 Color (APHA Pt-Co Scale). The color shall be determined as described in ASTM D 268-74 and ASTM D 2108-71. To be acceptable, the color shall meet the requirements specified in Table I.

4.4.1.8 Suspended matter.

4.4.1.8.1 Apparatus. A clean, dust-free, colorless, glass vessel of 100 ml volume shall be used.

4.4.1.8.2 Procedures. Tests shall be performed by adding approximately 100 milliliters of the sample to the vessel and examining the liquid while held between the light source and the observer. Note the presence of suspended matter as follows:

(a) Colloidal suspensions which cause the sample to be turbid or hazy.

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(b) Fine specks found floating in the sample.

(c) Particles that move toward the bottom of the sample. (The heavier particles will settle rapidly.)

To be acceptable, there shall be no more than five particles, specks, or combinations thereof of suspended matter or visible turbidity to meet the requirements of "substantially free" as specified in Table I.

4.4.1.9 Organic impurities.

4.4.1.9.1 Apparatus. The following apparatus shall be used:

(a) Gas chromatograph (preferably temperature programmed) with flame ionization or thermal conductivity detector and recorder

(b) Column, 3.2 millimeters by 1.8 meters stainless steel tubing, packed with Poropak Q, 100/120 mesh.

4.4.1.9.2 Conditions. The following conditions shall be obtained:

(a) Injection port - 200°C

(b) Detector - 200°C

(c) Column:

(1) Initial 120°C

(2) After 4 minutes, program a rise of 4°C per minute

(3) Maximum temperature 200°C.

NOTE: These readings are approximate and will vary with different instruments.

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4.4.1.9.3 Retention times. Retention times relative to propylene oxide are shown in Table II.

TABLE II. Retention times.

Compound	Relative retention time
Vinyl chloride	0.33
Acetaldehyde	0.53
Ethylene oxide	0.59
Propylene oxide	1.00
Propionaldehyde	1.12
Vinylidene chloride	1.18
1,2-dichloropropane	2.57
Propylene glycol	2.57

4.4.1.9.4 Calculations. Calculations shall be as follows:

(a) Measure the peak area of each compound. Correct the peak area of each compound by the relative response factor previously determined from pure compounds. Multiply the corrected peak area by 100 then divide by the sum of the corrected areas of all compounds. The result will be the mass percent of the compound. To be acceptable, the organic impurities content shall be no greater than the value specified in Table I.

4.4.1.10 Aldehydes (as propionaldehyde).

4.4.1.10.1 Reagents. Reagents shall be as follows:

(a) Sodium metabisulfite approximately 0.1N.

(1) Prepare sodium metabisulfite solution fresh daily as follows. Dissolve 4.9 to 5.0 gm of sodium metabisulfite, analytical reagent, in distilled water and make to 1 liter. Adjust the pH of the solution to 3.5 by careful addition of 0.2N sulfuric acid.

(b) Iodine solution (Potassium tri-iodide) 0.1N standardized.

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(c) Sulfuric acid, approximately 0.2N.

4.4.1.10.2 Apparatus. Apparatus used shall be as follows:

(a) Six 500 ml glass-stoppered Erlenmeyer flasks

(b) Magnetic stirrer and a PTFE-coated stirring bar

(c) Ice bath

(d) Iodine flask

(e) Burette, 25 ml.

4.4.1.10.3 Procedure. Tests shall be performed as follows:

(a) From a graduated cylinder, transfer 25 ml of ice-cold distilled water to each of six 500-ml Erlenmeyer flasks. Cool the flasks to 0°C to 5°C by immersing in an ice bath for approximately 5 minutes. Add carefully, by pipette, 20 ml of sodium metabisulfite solution to each flask. During this addition, immerse the tip of the pipette just beneath the surface of the liquid in the flask. Stopper the flasks and store by immersing in an ice bath from 5 minutes to 1.5 hours.

NOTE: Six flasks prepared at one time may be conveniently completed within a safe allotted time. Stoppered flasks containing only distilled water and sodium metabisulfite solution have been reserved in the ice bath for as long as 1.5 hours without any adverse effect.

(b) Remove a flask from the ice bath and add 50 ml of propylene oxide from a graduated cylinder. Immediately stopper the flask, swirl the contents, and let stand at room temperature for 15 to 20 minutes, swirling periodically. Chill the solution in an ice bath for 10 minutes.

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(c) Add to the flask, from a graduated cylinder, approximately 100 ml of ice-cold distilled water, and gently slide a PTFE-coated stirring bar into the flask. Set the flask on a magnetic stirring motor, and stir the solution at a rate just sufficient to form a vortex without throwing air bubbles into the solution. While stirring the solution, quickly add 15 ml of standard iodine solution from the burette.

NOTE: If the solution turns yellow, too large a sample is indicated. In this case, repeat the determination with a smaller sample.

Continue the addition of standard iodine solution until a yellow end point is produced that is permanent for at least 15 seconds. The total time for the addition of iodine should not exceed 5 minutes.

(d) Remove a flask from the ice bath storage and let stand at room temperature for 15 to 20 minutes. Recool it as above and titrate. At least three blank determinations should be made.

(e) To a 500-ml Erlenmeyer flask, add 225 ml of ice-cold distilled water and 50 ml of propylene oxide sample. While stirring the solution, titrate with standard iodine burette until a yellow end point is produced that is permanent for at least 15 seconds.

4.4.1.10.4 Calculation. Calculate aldehyde as follows:

$$\text{Percent aldehyde as propionaldehyde} = \frac{(A - B - C) \times N \times 2.9}{V \times (\text{specific gravity of sample})}$$

where:

A = average volume (milliliters) of iodine required for reagent blank

B = volume (milliliters) of iodine required for sample

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C = volume (milliliters) of iodine required for sample blank

N = normality of iodine

V = volume (milliliters) of propylene oxide sample.

To be acceptable, the aldehyde shall be not greater than the value specified in Table I.

4.4.2 Packaging, packing, and marking. Examination shall be made to ascertain that packaging, packing, and marking are in accordance with Section 5.

5. PACKAGING.

5.1 Preservation-packaging. Preservation and packaging shall be Level C. Propylene oxide, in the quantities specified in the contract or purchase order, shall be packaged in rust- and impurity-free containers in accordance with Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

5.2 Packing. Not applicable.

5.3 Marking. In addition to any special marking required by the contract (see 6.2), intermediate and exterior containers shall be marked in accordance with MIL-STD-129 and the Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

6. NOTES.

6.1 Intended use. This compound is intended for use with selected weapon systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

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(b) First article. When a first article is required, it will be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulations. The first article should be a preproduction sample. The first article should consist of not less than 1 kilogram of propylene oxide. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

(c) Testing activity designated to receive preproduction sample (see 3.2).

(d) Responsibility for performance of test requirements and test facilities if different than 4.1.

(e) Submittal of sample for testing (see 4.3.2).

(f) Special marking (see 5.3).

6.3 Definitions.

6.3.1 Batch. A batch is defined as that quantity of propylene oxide which has been manufactured by the same chemical process and subjected to the same physical mixing process intended to make the final product.

6.4 Explosibility. Propylene oxide is stable to detonating agents, but the vapor will undergo decomposition. Pure propylene oxide will decompose partially; however, a slight dilution with air or a small increase in initial pressure provides an ideal condition for complete decomposition. Copper or other acetylide-forming metals such as silver, magnesium, and alloys of such metals should not be used to handle or store propylene oxide because of the danger of the possible presence of acetylene. Acetylides detonate readily and will initiate explosive decomposition of propylene oxide vapor. In the presence of certain catalysts, liquid propylene oxide forms a poly-condensate. This reaction is highly exothermic (releases large amounts of heat). If the heat of the reaction is not conducted through the walls of a closed container at a rate capable of maintaining an equilibrium temperature, an increase in pressure results with an increase in reaction rate, leading to explosive conditions. Acid salts, such as stannic chloride and zinc chloride, and bases, such as alkali

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metal hydroxides, either solid or in aqueous solution, and tertiary amines are all effective catalysts. It is, therefore, imperative that the concentration of such contaminants be kept at a minimum when transporting or storing sizeable quantities of propylene oxide.

6.5 Toxicological. Liquid propylene oxide, concentrated or diluted, when exposed to the skin, can cause severe delayed burns. Short exposures produce mild first degree burns, but prolonged exposures produce second degree burns with the formation of large blisters. Exposure to the vapor results in systemic manifestations and irritation to the respiratory system. Inhalation of propylene oxide vapors, if prolonged, results in severe systemic (affecting the body generally) poisoning with the symptoms of nausea, vomiting, headache, dizziness, and diarrhea.

6.6 Safety precautions. Propylene oxide is a volatile, flammable, and reactive chemical. The liquid is reactive, and the vapors form flammable or explosive mixtures with air. Both liquid and vapors are toxic to personnel; therefore, propylene oxide should be handled with extreme care. Keep away from heat, sparks, and fire. Do not breathe the vapor. Do not leave the container open and avoid contact with eyes or skin. In case of contact with eyes or skin, flush immediately with water for at least 15 minutes; for eyes, get immediate attention.

Preparing activity:
Navy - AS

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